MILITARY SPECIFICATION

CORRECTOR, WIND SOUND RANGING: M1A1

1. SCOPE

1.1 This specification covers Sound Ranging Wind Corrector M1A1 which is used to solve wind corrections graphically (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

F10511095	Corrector, Sound Ranging Wind: M1A1

PACKAGING DATA SHEET

10511095	Packaging of Corrector, Sound Ranging Wind: M1A1

FSC 1220

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(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The wind corrector shall be manufactured in accordance with Drawing F10511095 and drawings pertaining thereto.

3.2 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.3 Environmental.

3.3.1 Storage temperatures.- The wind corrector shall show no evidence of physical failure and shall meet all performance requirements of this specification at a standard ambient temperature of between 60° and 90°F. after having been exposed to and thermally stabilized at ambient temperatures of -80° and +160°F.

3.3.2 Operating temperatures.- The wind corrector shall meet the applicable requirements of 3.5 while exposed to and thermally stabilized at ambient temperatures of -40° and +150°F.

3.4 Accuracy.

3.4.1 Alignment of scales.- The inner and outer graduation lines for the azimuth values listed in Table I shall be in alignment within one-half the width of a graduation line.

TABLE I

Inner Azimuth Scale Graduation (Mils)	Outer Azimuth Scale Graduation (Mils)
0	0
800	45
1600	90
2400	135
3200	180
4000	225
4800	270
5600	315

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3.4.2 Correction readings.- With the index of the correction scale in coincidence with the inner azimuth scale graduations listed in Table II, and with the wind velocity arm rotated to the azimuth settings noted, the correction readings indicated in the table shall be obtained for the wind velocities listed. The full width of the outer azimuth scale graduation lines (.010 + .005) shall be utilized in setting the wind velocity arm. This tolerance shall permit the exact wind correction readings listed.

TABLE II

Wind Velocity Arm Azimuth Setting (Outer Azimuth Scale)	Correction Scale Azimuth Setting (Inner Azimuth Scale)	Wind Velocity (On Wind Velocity Arm Scale)	Wind Correction Scale Reading
50°	400 mils	13 knots	-.07 sec.
50°	400 mils	26 knots	-.14 sec.
130°	6000 mils	13 knots	+.07 sec.
130°	6000 mils	26 knots	+.14 sec.
130°	2800 mils	13 knots	-.07 sec.
130°	2800 mils	26 knots	-.14 sec.
230°	400 mils	13 knots	+.07 sec.
230°	400 mils	26 knots	+.14 sec.
230°	3600 mils	13 knots	-.07 sec.
230°	3600 mils	26 knots	-.14 sec.
310°	6000 mils	13 knots	-.07 sec.
310°	6000 mils	26 knots	-.14 sec.

3.5 Operability.- The wind velocity arm and correction scale disk shall rotate freely. The subbase azimuth scale indices shall slide within the annular groove between the inner and outer azimuth scales without binding or excessive looseness. The clamp and clamping screw located at the end of the wind velocity arm shall be capable of locking the arm in any position.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

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4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 wind correctors manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAPs, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables III, IV and V herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE III

CLASSIFICATION CRITICAL: NONE MAJOR: AQL 1.0% Defective	REQUIREMENT	TEST PROCEDURE
101. Alignment of scales	3.4.1	4.6.3
102. Correction readings	3.4.2	4.6.4
103. Operability	3.5	4.6.5
MINOR: NONE		

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NOTE 1: The tests in Table III shall be conducted at a standard ambient temperature of between 60° and 90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- Three wind correctors shall be selected at random as a special sampling from each 100 produced. The samples shall meet the requirements and tests in Table IV.

TABLE IV

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Drawings - Visual
302.	General specifications	3.2	MIL-F-13926-Visual

NOTE 2: The tests in Table IV shall be conducted at a standard ambient temperature of between 60° and 90°F.

4.4.2 Environmental tests.- Three wind correctors shall be selected at random from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table V and shall then meet the requirements in Table III.

TABLE V

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperatures	3.3.1	4.6.1
304.	Low temperature operation (-40°F)	3.3.2	4.6.2
305.	High temperature operation (+150°F)	3.3.2	4.6.2

4.4.3 Failure of sample.- Should any item of a special sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

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4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the Technical Data Package (TDP).

4.5.2.2 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Storage temperature.- The wind correctors shall be exposed to the temperature conditions specified in 3.3.1, utilizing testing equipment in accordance with the "Test Facilities" requirements of MIL-F-13926. The wind corrector shall be placed in the test chamber and the temperature of the chamber reduced gradually (see NOTE 3) to -80°F and shall remain at this temperature for a minimum of one hour. At the completion of this one hour period, the temperature shall then be gradually raised to +160°F and the wind correctors shall remain at this temperature for a minimum of one hour. At the completion of this one hour period, the temperature of the test chamber shall be gradually reduced to room ambient temperature (+60° to +90°F) at which time the wind correctors shall be removed from the test chamber. The wind correctors shall then be subjected to a visual and tactile examination to determine that there is no evidence of physical failure. The wind correctors shall then be subjected to all the tests specified herein to determine overall compliance with 3.3.1. Caution shall be exercised during this test to avoid subjecting the wind correctors to thermal shock.

NOTE: The rate of temperature change in the test chamber shall not exceed 20°F per hour throughout the temperature cycling tests of 4.6.1 and 4.6.2.

4.6.2 Operating temperature.- The wind correctors shall be exposed to the temperature conditions specified in 3.3.1, utilizing testing equipment

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in accordance with the "Test Facilities" requirements of MIL-F-13926. The wind correctors shall be placed in the test chamber and the temperature of the chamber reduced gradually (see NOTE 3) to -65°F. The wind correctors shall remain at this temperature for a minimum on one hour. At the completion of this one hour period, and while at -65°F the wind correctors shall be subjected to the test in 4.6.5. Upon completion of this test, the temperature of the test chamber shall be raised gradually to +150°F. The wind correctors shall remain at this temperature for a period of one hour. At the completion of this one hour period, and while at +150°F the wind correctors shall again be subjected to the test in 4.6.5. Upon completion of this test, the temperature of the test chamber shall be gradually reduced to room ambient temperature (+60° to +90°F) at which time the wind correctors shall be removed from the test chamber. The wind correctors shall conform to the requirements of 3.3.2. Caution shall be exercised during this test to avoid subjecting the wind correctors to thermal shock.

4.6.3 Alignment of scales.- The alignment of the graduation lines between the inner and outer azimuth scales as specified in 3.4.1 shall be determined by means of a commercial type optical comparator with a magnification of at least 5 power. The comparator shall contain an appropriate reticle pattern to facilitate in the measuring for any misalignment between the graduation lines of the scales at the settings specified in Table I.

4.6.4 Correction readings.- Utilizing an optical comparator as in 4.6.3, the corrector shall be subjected to a tactile-visual examination to determine the accuracy of the correction readings. Utilizing the settings in Table II, set the wind velocity arm to a selected azimuth setting in degrees. Coincide the correction scale index to the selected azimuth setting (in mils). Determine the wind correction scale reading for the prescribed wind velocity listed in Table II. The wind correction scale reading shall be obtained within the adjustment limits specified in 3.4.2.

4.6.5 Operability.- Tactile inspection technique shall be used to determine conformance with the requirements of 3.5. With normal finger pressure applied, the wind velocity arm and the correction scale disk shall be moved back and forth several times and then through their full mode of travel to determine conformance with the pertinent requirement of 3.5. Each of the subbase indices shall slide without bind or looseness within the groove when normal finger pressure is applied. The clamping assembly shall provide positive locking action when the clamping screw is drawn up snug. The clamp and clamping screw assembly shall be tested for positive locking action in each of the four quadrants.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10511095. The level of protection shall be as specified in the procurement document.

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6. NOTES

6.1 Intended use.- The Sound Ranging Wind Corrector M1A1 determines required corrections for wind velocity and direction. It is used in conjunction with Sound Ranging Plotting Boards M1 and M1A1, and the components of Sound Ranging Plotting Set M53 in plotting the location of hostile batteries.

6.2 Supersession data.- Specification MIL-W-12799A(MU) was cancelled as of July 1964. It is being reinstated as MIL-C-12799B(MU) with updated requirements.

6.3 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Selection of applicable levels of preservation, packaging and packing
(c) Applicable packaging data sheet number (see 5.1)
(d) Applicable stock number
(e) Provisions for First Article testing

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1220- A226

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERE. DO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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